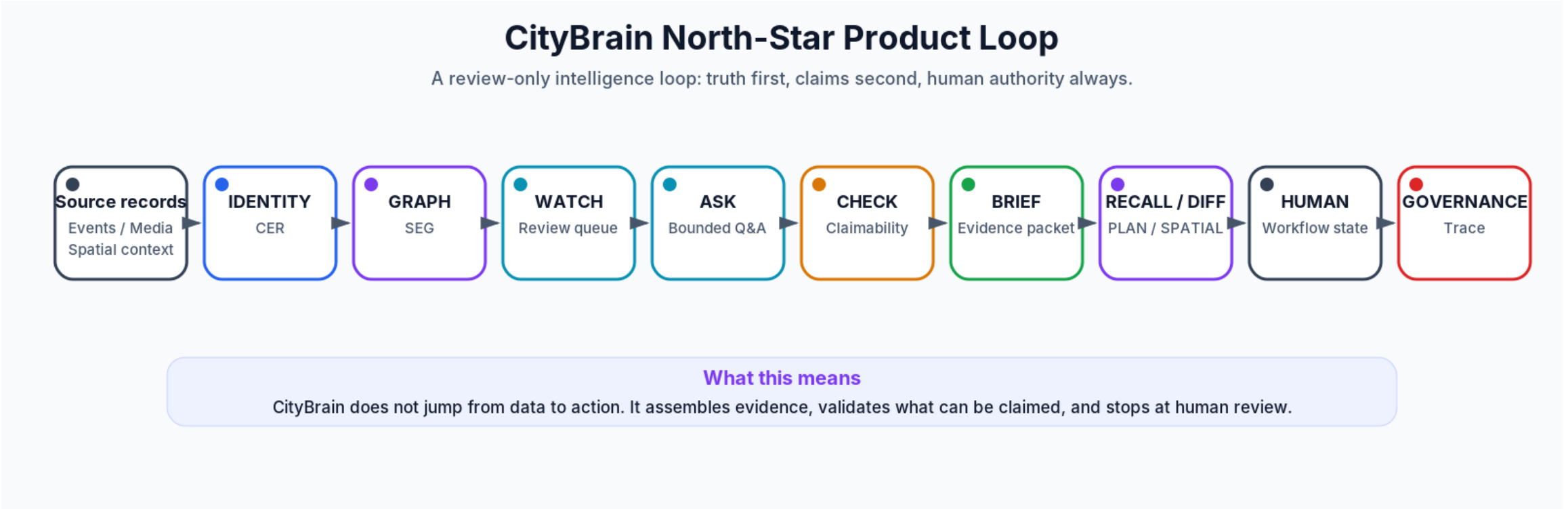


CityBrain Product Consolidation

Product, architecture, gaps, parked work, and component plug-in map



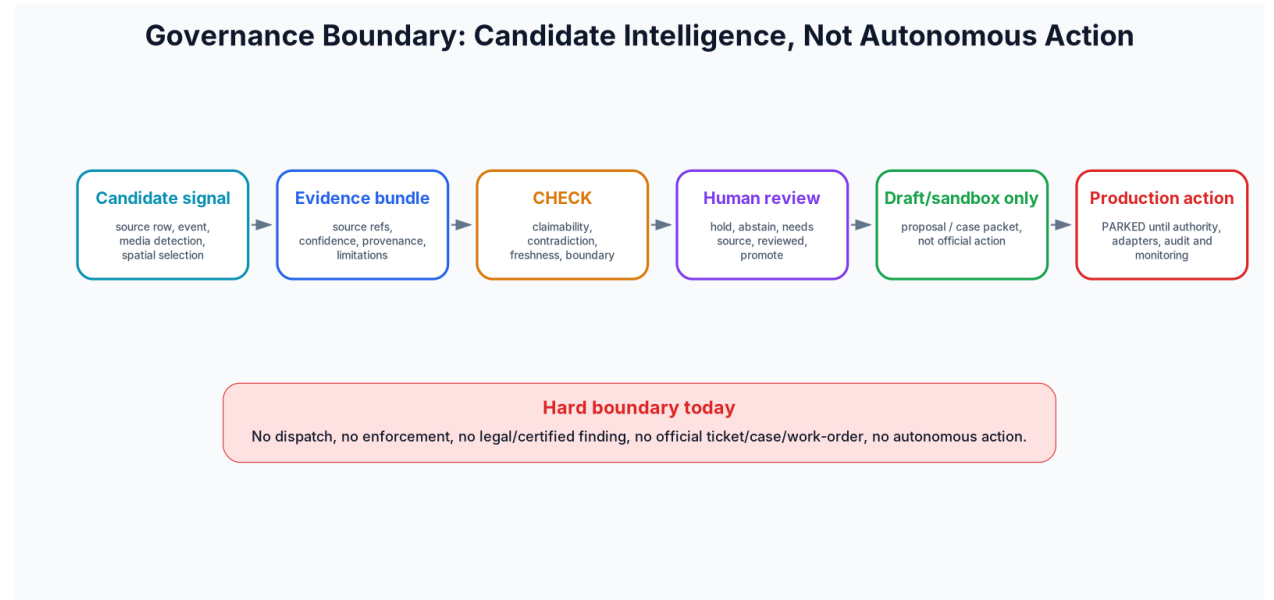
The product truth

CityBrain is a governed review-only intelligence cockpit.

CityBrain is a governed review-only city intelligence cockpit that turns fragmented city records, events, media candidates and spatial context into evidence-backed human-review packets.

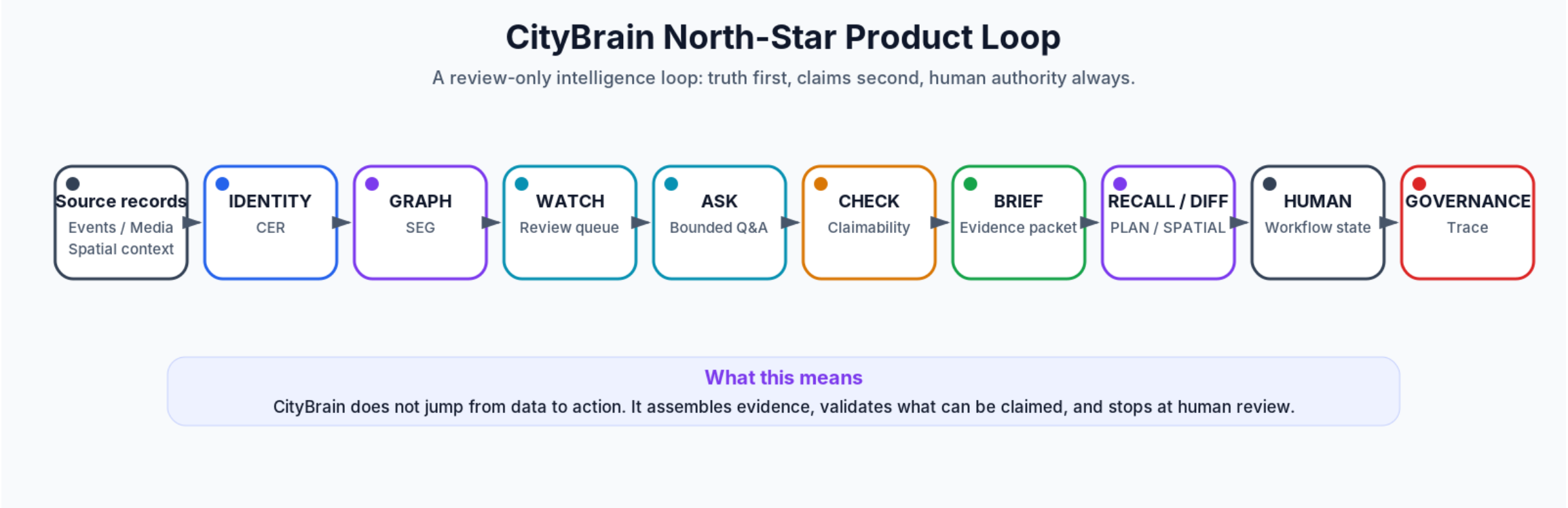
Current strength: trusted local/replay review packets, not autonomous operations.

The product unit is a packet: entity + graph + evidence + CHECK + authority + workflow state.



North-star loop

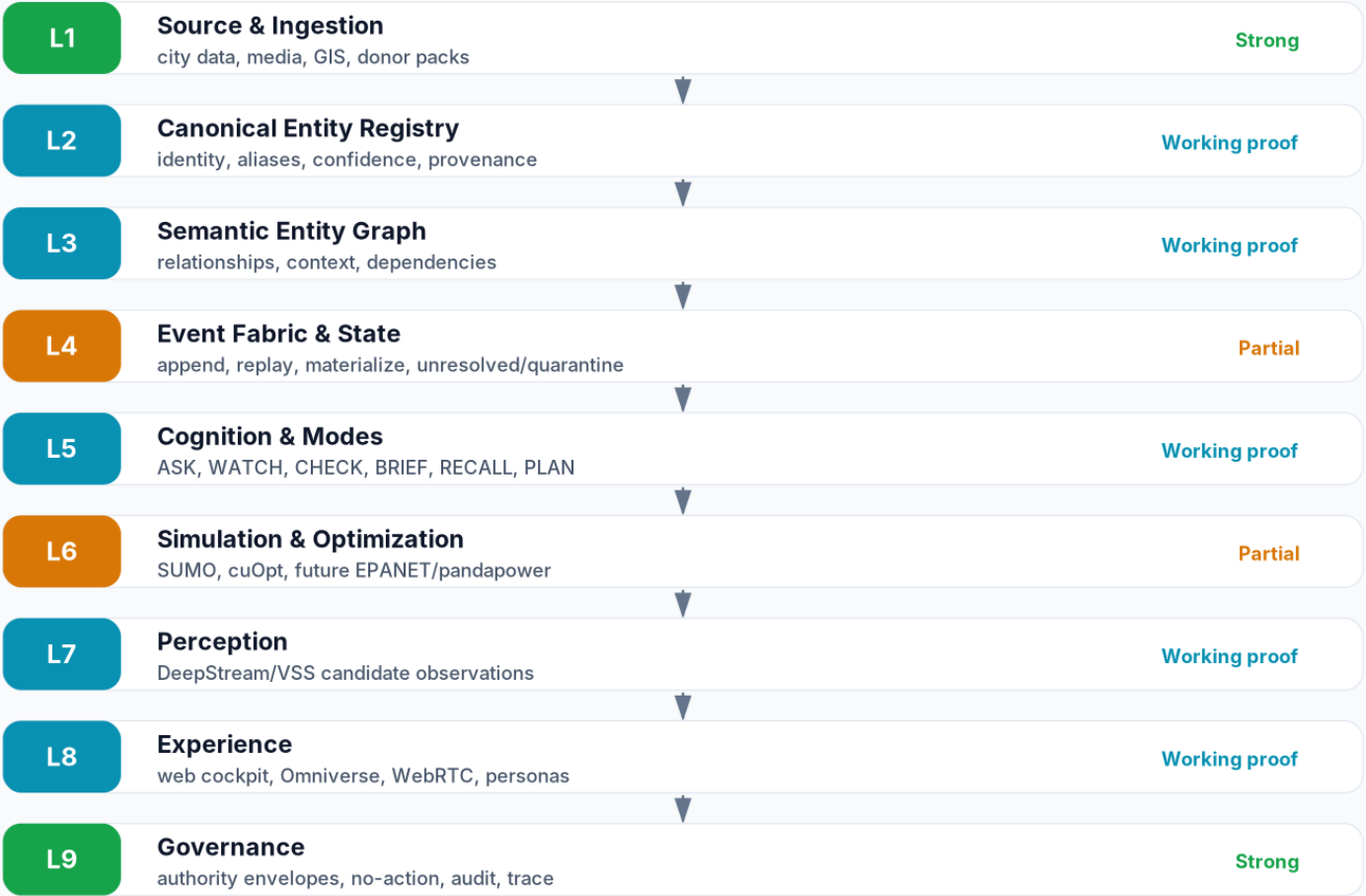
Truth flows through identity, graph, evidence, CHECK, human review and governance.



Architecture spine

CER first, graph second, evidence and governance always

Architecture Spine: What Exists vs What Still Needs Depth

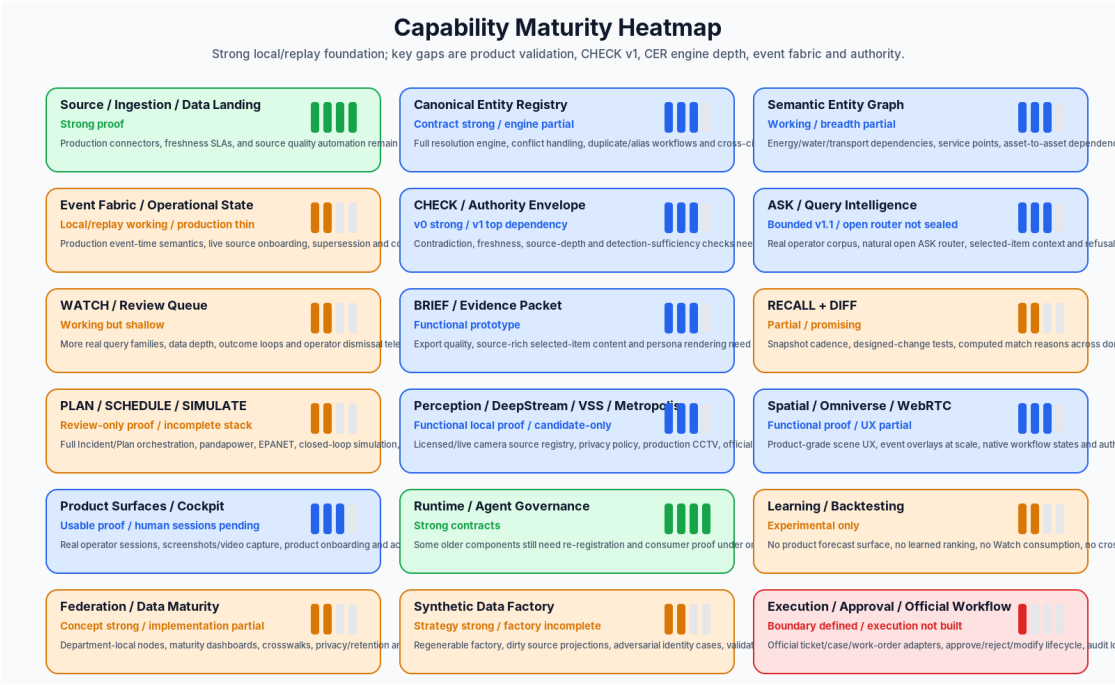


Key correction: this is not a production command system. It is a local/replay governed review platform with several strong proof paths.

What exists today

Strong proof paths, but conservative claims.

- Source records and data landing across several cities.
- Canonical graph and evidence packet proofs.
- Web cockpit, Omniverse/WebRTC and DeepStream/VSS local proof lanes.
- CHECK/Authority/trace/no-action governance patterns.
- One offline experimental forecast, frozen-replay only.



What cannot be claimed yet

This honesty is a credibility feature.

Production live operations.

Public API / SLA.

Certified citywide digital twin.

Autonomous monitoring or dispatch.

Enforcement/legal/certified finding.

Official ticket/case/work-order creation.

Product forecast surface.

Live Watch consumption of a forecast model.

Cross-city learned transfer.

Trained ranking in operator surfaces.

Production-grade execution adapters.

Official execution remains parked until authority exists.

Product management map

Different users see the same evidence chain through different lenses.

Product Management Map: User Value by Audience

Operator

What needs review, what can I safely do next?

WATCH, CHECK, selected-item workspace, workflow state, safe next looks

Analyst / planner

What do we know, what changed, what is similar?

ASK, RECALL, DIFF, graph context, source-record 360

Executive / client

What is the story, value and risk boundary?

BRIEF, option-set summary, risk report, demo route

Data / twin engineer

Where is truth, geometry and identity?

Source registry, CER/SEG, spatial binding, data maturity

Governance / risk

What must not be claimed or executed?

CHECK, AuthorityEnvelope, no-action ledger, guardrails

ML / evaluation

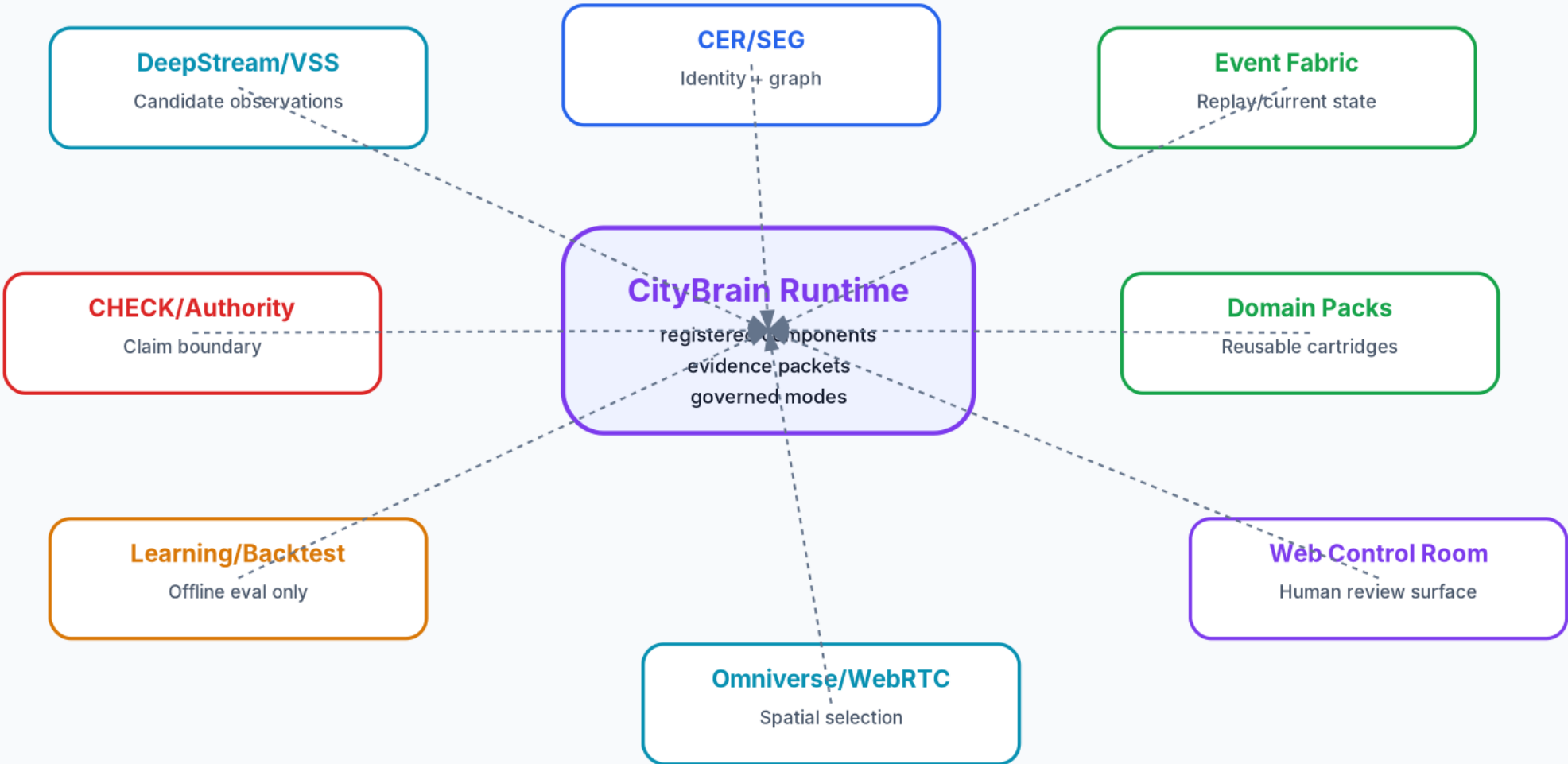
What can be learned safely?

Outcome ledger, calibration, backtest harness, experimental forecast registry

Component plug-in map

The I

Scattered Components: Where They Plug In Now



CityBrain - governed review intelligence, not production command/control

Gap priority matrix

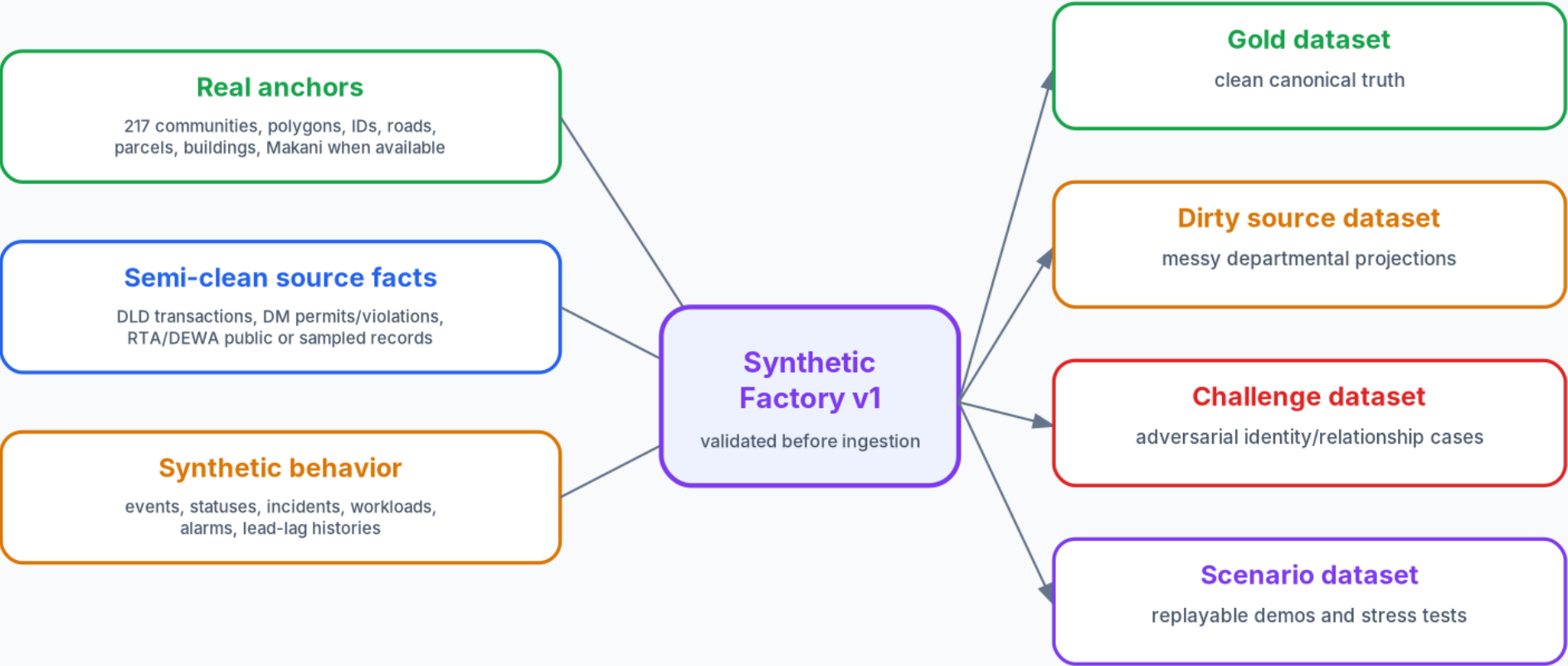
P0 gaps are



Synthetic and Dubai strategy

Real

Dubai / Synthetic Data Strategy



The demo should be real where identity matters and synthetic where operational behavior is missing.

CityBrain - governed review intelligence, not production command/control

Scenario portfolio

One spine, many cartridges

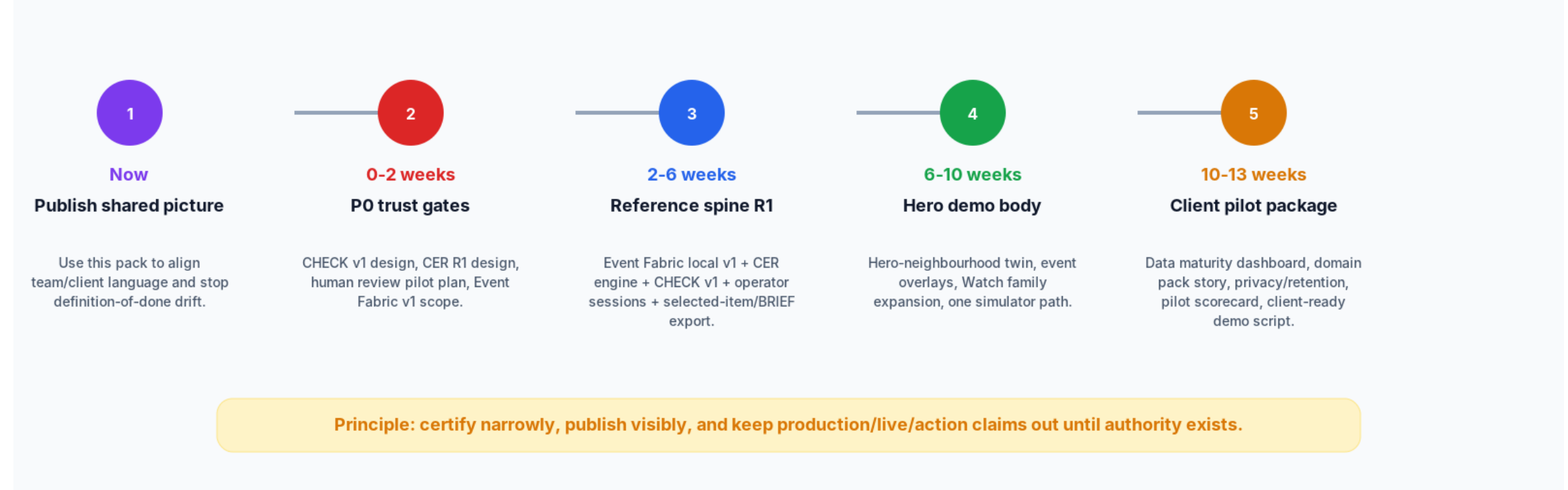
Scenario / Flow Portfolio

Flow 1	Situational status Descriptive · Donor: NYC/Dubai · Cheap next cartridge; status of area X.
Flow 2	Construction compliance cascade Reactive / golden path · Donor: Dubai + NYC · Deep thread: vision -> graph -> evidence -> option/brief.
Flow 3	Resilient city: fire + road + air Reactive · Donor: London · Not yet proven as Flow 3; London foundation exists.
Flow 4	Crowd surge / major event Proactive · Donor: Melbourne/Singapore · Needs LTA/key and event/crowd data depth.
Flow 5	Flood / asset dependency cascade Proactive / forecast · Donor: NYC/London/ERA5 · Requires flood/weather/water/energy simulation.
Flow 6	Planned shutdown sequencing Proactive / domain swap · Donor: Oil & gas cartridge · Powerful story but separate domain pack.
Flow 7	Civic service + sensor fusion Reactive · Donor: Chicago · Strong flat-file/civic proof candidate.

Roadmap

Consolidate trust foundation, then demo/pilot.

Roadmap: From Consolidation to Client Pilot



Immediate action plan

Make the picture real, testable and explainable.

- Publish this pack as the single product room.
- Build CHECK v1 and CER Engine R1.
- Run first real operator sessions.
- Finish local Event Fabric v1.
- Build hero-neighbourhood twin with event overlays.
- Package a client demo route with strict claims.

Client close

The safest, strongest message.

CityBrain is a review-and-governance layer between messy city reality and eventual approved operations.

It helps humans know what happened, what matters, what can be claimed, what should be checked next and what is still forbidden.

The current product is trusted review, not autonomous command.